

Cooper Kimball-Rhines, Ph.D.

Data Science, Policy, Conservation

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I work with ecosystem managers and decision-makers to improve plant biodiversity conservation. My research is collaborative and interdisciplinary with a focus on implementing genetic, geospatial, and causal modeling into conservation planning and policy.

Education

Ph.D., Environmental Biology, 2026

University of Massachusetts Boston

NSF Graduate Research Fellow

Dissertation: Adapting population genetics, practice, and policy for plant conservation in a changing world; Advisor: Brook Moyers

Committee: Mayra C. Vidal, Georgia Mavrommati, Alex Harkess, Bob F. Chen

B.Sc., Microbiology and Immunology, 2021

McGill University, School of Biomedical Sciences

Dean's Multidisciplinary Research List

Thesis: The PFAS package in *enviPath*: a resource for the prediction of AFFF biotransformation; Advisor: Jinxia Liu

Work Experience

Research Fellow, COBALT, June 2026-present

Geospatial data analyst for seagrass restoration planning in Casco Bay, Maine

Developing R Shiny tools to visualize field data for decision-making

Supporting data integration into digital twin framework

Research Fellow, University of Massachusetts Amherst, June 2026-present

Facilitating USGS NECASC RISCC working group on climate-smart restorations

Synthesizing knowledge on grassland plant genetics for restoration planning

Supervisor/Advisor: Bethany Bradley

Pathways Intern, United States Environmental Protection Agency, Oct. 2021-Aug. 2024

EPA Student Intern of the Year 2023

Region 1 Superfund and Emergency Management Division

Assisted on-scene coordinators in assessing contamination at emergency removals

Manager: Stacy Greendlinger

Roadside Pollinator Assistant, New Hampshire Fish and Game, May-Sept. 2019-2021

Performed monarch butterfly surveys to inform NH DOT policy on habitat management

Supported Karner Blue Butterfly, Frosted Elfin, and White Mountain Fritillary projects

Manager: Heidi Holman

Publications

6. **Kimball-Rhines, C.**, Mello E., Taveras Guzman, S., Moyers B. Divergence in genotype and methylation are decoupled and are not driven by nitrogen source in a salt marsh plant. In prep for *Evolutionary Applications*
5. **Kimball-Rhines, C.**, Taveras Guzman, S., Mavrommati, G., Moyers, B. Leaping a conservation genetics gap: A collaborative, multidisciplinary approach to restoration genetics in the pine barren plant *Lupinus perennis*. In prep for *Conservation Science and Practice*
4. Keller, AG., **Kimball-Rhines, C.**, Leng, D., McIntosh, T., Nelson, K., Tension between opportunities and risks of deep learning in social-ecological systems research. In review, *Ecology and Society*
3. **Kimball-Rhines, C.**, Moyers, B. Missing the forest for the trees: A review and case study of state conservation law. Accepted, *Oryx*.
2. Comerford, M.S., Nash, K., Germaine, A.S., Andrews, C., Barr, T.S., Dain, J., Gonclaves, A., Jones, A., **Kimball-Rhines, C.**, Maguire, W., Richards, A., Schneider, E., Wilkes, A. and Vidal, M.C. (2026), Quantitative synthesis of the effects of drought on community composition and species interactions in terrestrial ecosystems. *Oikos* e12026.
<https://doi.org/10.1002/oik.12026>
1. **Cooper Kimball-Rhines**, Alice E Kaisla, Scarlet Taveras Guzman, Emily Mello, Karol Vanessa Rojas Ramirez, Alex Harkess, Brook Moyers, The annotated, chromosome-scale *Salicornia depressa* (American pickleweed) genome, *Journal of Heredity*, 2025; esaf062,
<https://doi.org/10.1093/jhered/esaf062>

Research Grants and Fellowships

2023-2028	NSF Graduate Research Fellowship	\$159,000
2023-2025	Nantucket Biodiversity Initiative Climate Change and Island Resilience	\$5,506
2023-2025	DOE Joint Genome Institute Community Science Program	RNA-seq Services
2023	UMB Transdisciplinary Dissertation Proposal Development Program	\$3,000
2023	UMB Nancy Goranson Endowment Fund	\$735
2022	New England Botanical Society Graduate Student Research Award	\$1,500

Teaching Experience

Curriculum Maintainer, The Carpentries, Intro to R Geospatial Data, May 2025-present
Instructor, iTCGA Computation for Cancer Genomics Workshop, June 2-20, 2025
Instructor, The Command Line for Genomics Workshop, July 15-17, 2024
Instructor, GIS in R for Casco Bay Workshop, Jan. 10-12, 2024
Certified Instructor, The Carpentries, Dec. 2023-present
Teaching Assistant, UMass Boston, Biostatistics and Experimental Design, Sept-Dec. 2023

Organized Workshops

NECASC Climate Adapted Restoration Meeting, April 9, 2026, Online hosted by the
Northeast Regional Invasive Species and Climate Change Network, Amherst, MA

Invited Workshops

ESIIL	Leadership Program and Innovation Summit on AI for Sustainability: Translating Environmental Data into Decisions, May 11-14, 2026, Boulder, CO
AIBS	Policy Communication Bootcamp and Congressional Visits Day, April 28-30, 2025, Washington, DC
NCEAS	Environmental Data Science Summit: AI and Conservation Management, Feb. 4-6, 2025, Santa Barbara, CA

Media Features

2. Lambert, L. O. (2026, June 24) The Plight of the Karner Blue Butterfly. *National Wildlife Magazine*. Summer Issue. <https://www.nwf.org/Home/Magazines/National-Wildlife/2026/Summer/PhotoZone/Karner-Blue-Butterfly>
1. Brown, J. R. (2026) Vermont Lupine gets a Boost from Banked Seed. *Native Plant News*. Spring/Summer Issue. https://www.nativeplanttrust.org/documents/1821/NPT_loresSPREADS_NPNews.SprSum2026_28pg_1.pdf

Presentations

10. Poster: **Kimball-Rhines C.**, Taveras Guzman, S., Moyers B. "Gene Pool or Puddle: When 'Local is Best' Meets Restoration Reality." Ecological Society of America 2025; August, 2025, Baltimore, MD
9. Poster: **Kimball-Rhines C.**, Moyers B. "Divergence and Diversity Decoupled: A Population Methyloomic Approach." Harvard Plant Biology Initiative Meeting; May, 2025, Boston, MA.
8. Poster: **Kimball-Rhines C.** "Drowning to Diverse: A causal analysis of salt marsh sediment placement." Northeast Estuarine Research Society Meeting; April, 2025, Provincetown, MA.
7. Oral Presentation: **Kimball-Rhines C.**, Palmer A., Harkess A., Guzman-Taveras S., Mello E., Rojas V., Moyers B. "Six Hot Tips learned from assembling a tetraploid halophyte genome." Third Joint Congress on Evolutionary Biology; July, 2024, Montreal, QC, CA.
6. Oral Presentation: **Kimball-Rhines C.**, Palmer A., Moyers B. "Genetics for Marsh Conservation." Nantucket Biodiversity Initiative Meeting; November, 2023, Nantucket, MA.
5. Seminar: **Kimball-Rhines C.**, Palmer A., Moyers B. "Live Demonstration of DNA Sequencing for Marsh Conservation." Nantucket Field Station; August 15, 2023, Nantucket, MA.
4. Poster: **Kimball-Rhines C.**, Moyers B. "One Fish, Two Fish, Sea Fish, Bee Fish: Assessing State-Level Biodiversity Policy." Evolution in Small Population Meeting; July, 2023, Princeton, NJ.
3. Poster: **Kimball-Rhines C.**, Moyers B. "One Fish, Two Fish, Sea Fish, Bee Fish: Assessing State-Level Biodiversity Policy." Society for the Study of Evolution; June, 2023, Albuquerque, NM.
2. Poster: **Kimball-Rhines C.**, Palmer A., Moyers B. "Genome Profiling of *Salicornia depressa*." Ecological Society of America Meeting; August, 2023; Montreal, QC, CA.
1. Lightning Talk: **Kimball-Rhines C.**, "Genome Profiling of *Salicornia depressa*" UMass Boston Earth Day Symposium; March 2022; Boston, MA.